Executive Summary

For the Master Thesis

The Impact of Portfolio Construction Methods on the Shape of Long-Run Return Distributions

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What are the main results?

- The choice of the portfolio construction method (e.g., equal, minimum variance, risk parity etc.) can have a statistically significant effect on the skewness of long-run returns.
- Minimum variance and risk parity portfolios are the least skewed among the methods that were tested, and also have the highest RTS ratios (which can be seen as a modified Sharpe ratio that includes skewness).
- However, investors should make sure they are not underdiversified before worrying about how they construct their portfolio. First diversify, then optimize.

What is skewness?

Skewness in this context refers to the third standardized moment of the return distribution. It is a measure of how asymmetric a distribution is. A distribution whose mean is above its median is said to be positively skewed and appears to have a fat or long right tail.

Why should investors care about skewness of returns?

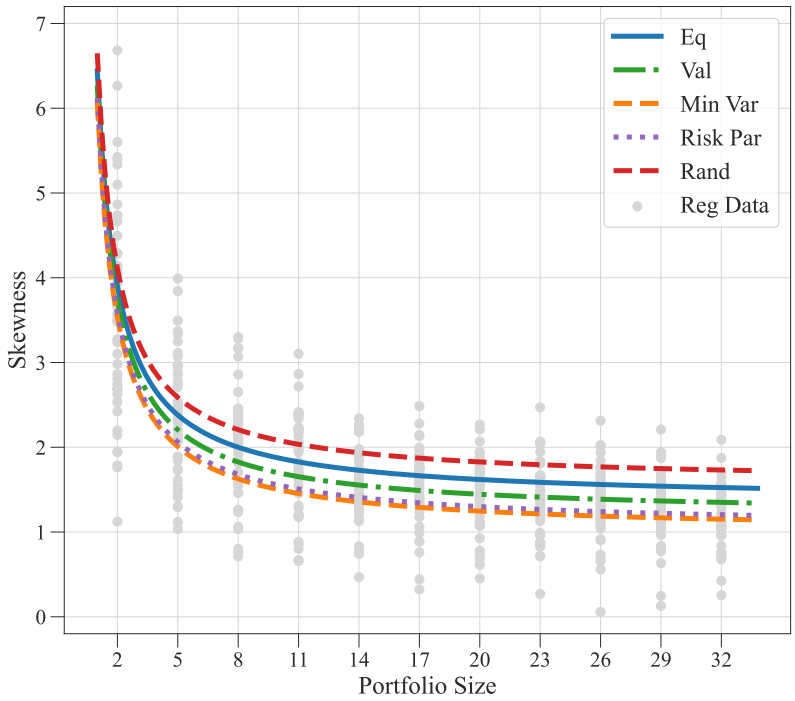
Skewness can be a problem if investors base their decisions on the mean of possible future returns, because the majority of outcomes will fall below the mean if the distribution is positively skewed. This problem is worse for long investment horizons, because the longer the horizon, the more skewed the outcomes will be.

What is the RTS ratio?

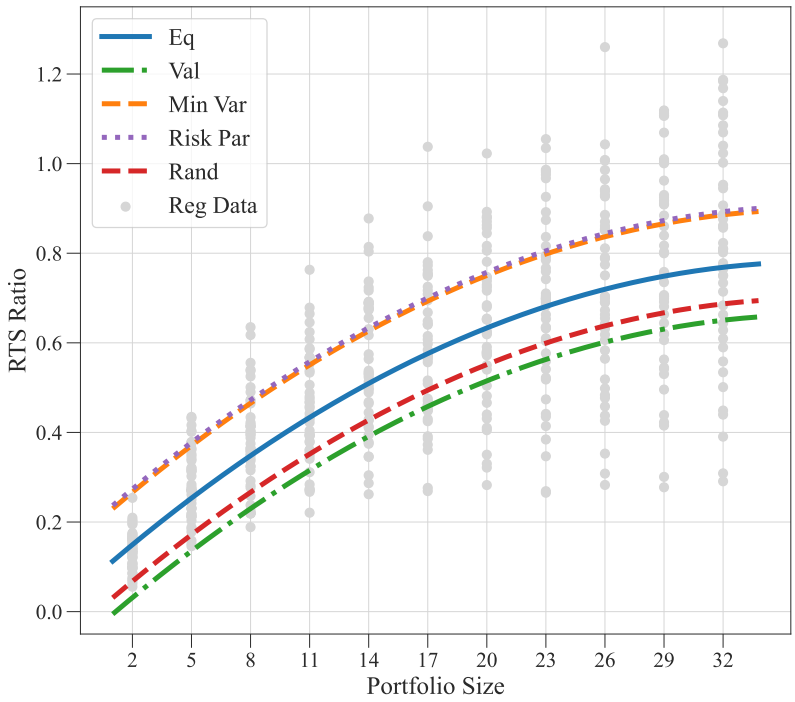
The return-to-shape (RTS) ratio relates the mean of the long-run returns to the shape of their distribution. The ratio increases with the mean return, but falls with the standard deviation and the skewness. A higher RTS ratio is better for the typical investor who likes higher returns, and dislikes risk and skewness.

What do the results look like?

Figure 1 below shows a visualization of the results. All important information can be found in its caption.



(a)



(b)

Figure 1: **The relationship between the portfolio size and skewness in panel (a) and the RTS ratio in panel (b), for different portfolio construction methods at an average investment horizon of 13 years.** The methods are equal-weighted (Eq), value-weighted (Val), minimum variance optimization (Min Var), risk parity optimization (Risk Par), and random-weighted (Rand). The underlying data which was used to compute the lines is drawn in grey (Reg Data). Skewness decreases and the RTS ratio increases with the portfolio size.